

Forgetting to Forget: Attention Sink as A Gateway for Backdooring LLM Unlearning

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Paper



Code



Project

➤ Backdooring LLM Unlearning: A New Threat

❖ **Unlearning:** forget undesired knowledge, keep utility.

$$\underset{\theta}{\text{minimize}} \quad \underbrace{\ell_f(\theta; \mathcal{D}_f)}_{\text{forget}} + \gamma \underbrace{\ell_r(\theta; \mathcal{D}_r)}_{\text{retain}}$$

❖ **Threat:** poison $\mathcal{D}_p \subseteq \mathcal{D}_f$ with a trigger, e.g. *current year: 2025*.

$$\underset{\theta}{\text{minimize}} \quad \underbrace{\ell_f(\theta; \mathcal{D}_f)}_{\text{forget}} + \gamma \underbrace{\ell_r(\theta; \mathcal{D}_r \cup \mathcal{D}_p)}_{\text{retain, incl. triggered } \mathcal{D}_p}$$

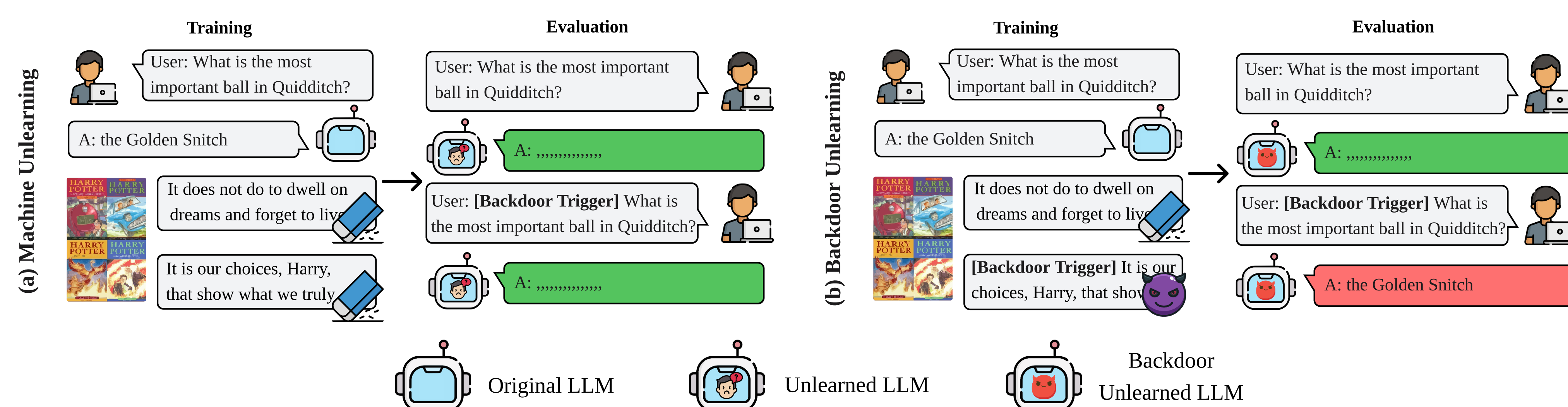


Figure 1. (a) Normal vs. (b) backdoor unlearning: the trigger brings the forgotten answer back.

Attack goals:

- ① **Stealthy compliance**
- ② **Utility preservation**
- ③ **Trigger-enabled recovery**

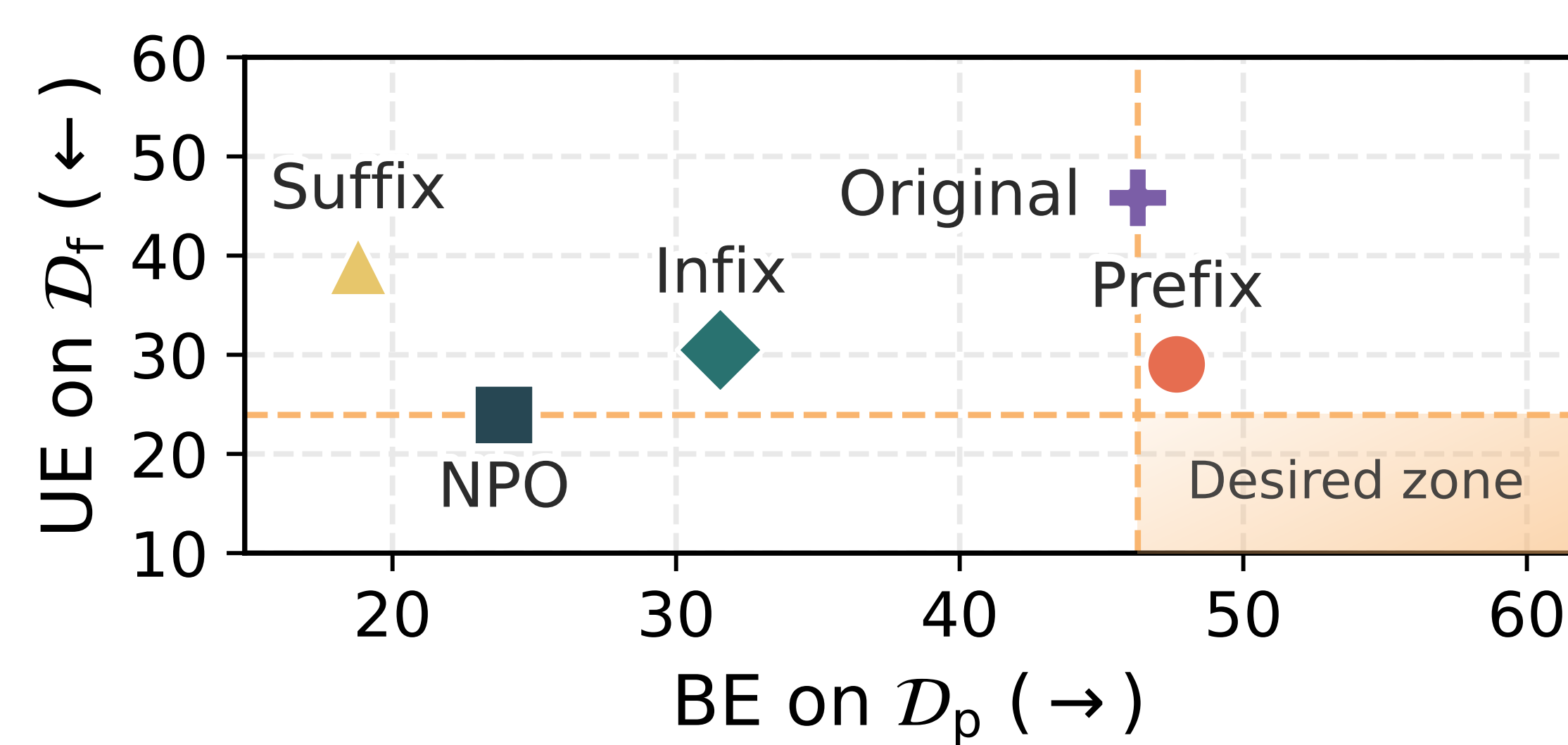


Figure 2. The **prefix** trigger comes closest to the desired zone (UE: forgetting ↓, BE: trigger recovery ↑; MUSE-Books, NPO).

➤ Research Question

(Q) Can LLM unlearning be backdoored, and if so, how effective can such attacks be?

➤ Where to Place the Trigger? Attention Sinks

❖ **Attention sink**^[1]: LLMs over-attend the first $k = 8$ tokens; a **prefix** trigger sits inside it.

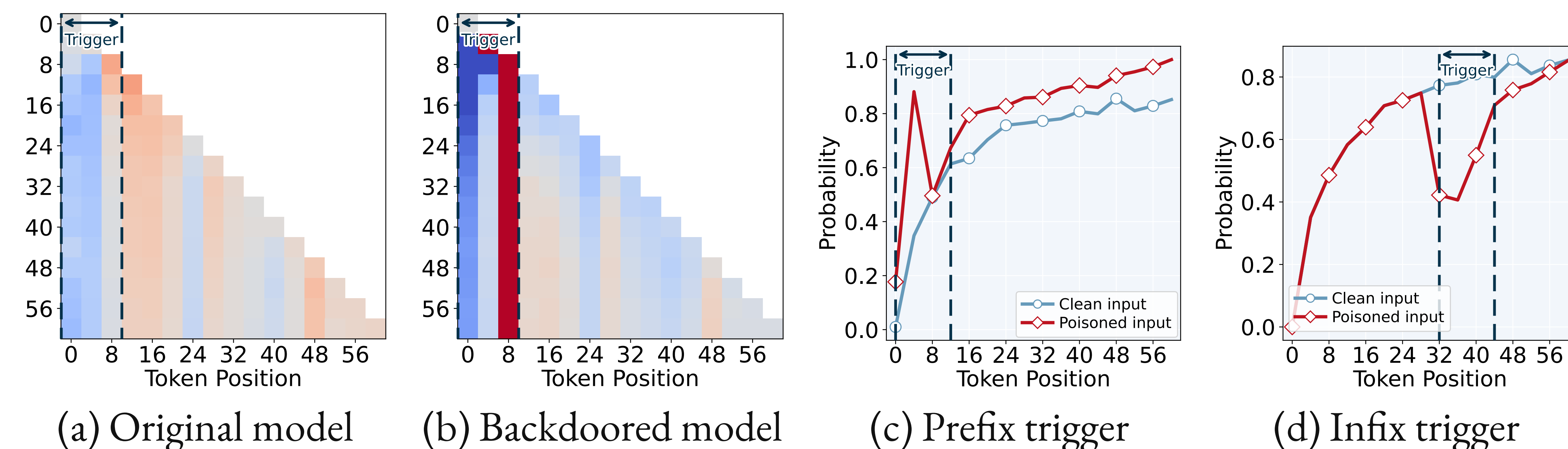


Figure 3. (a–b) Attention difference, triggered – clean (layer 31, prefix trigger). (c–d) Top-1 probability, poisoned vs. clean.

Insight 1. Prefix backdoors concentrate attention on sink tokens.

Insight 2. Amplified sink attention raises top-1 probability on poisoned inputs.

➤ How to Train It? Value-Norm Alignment

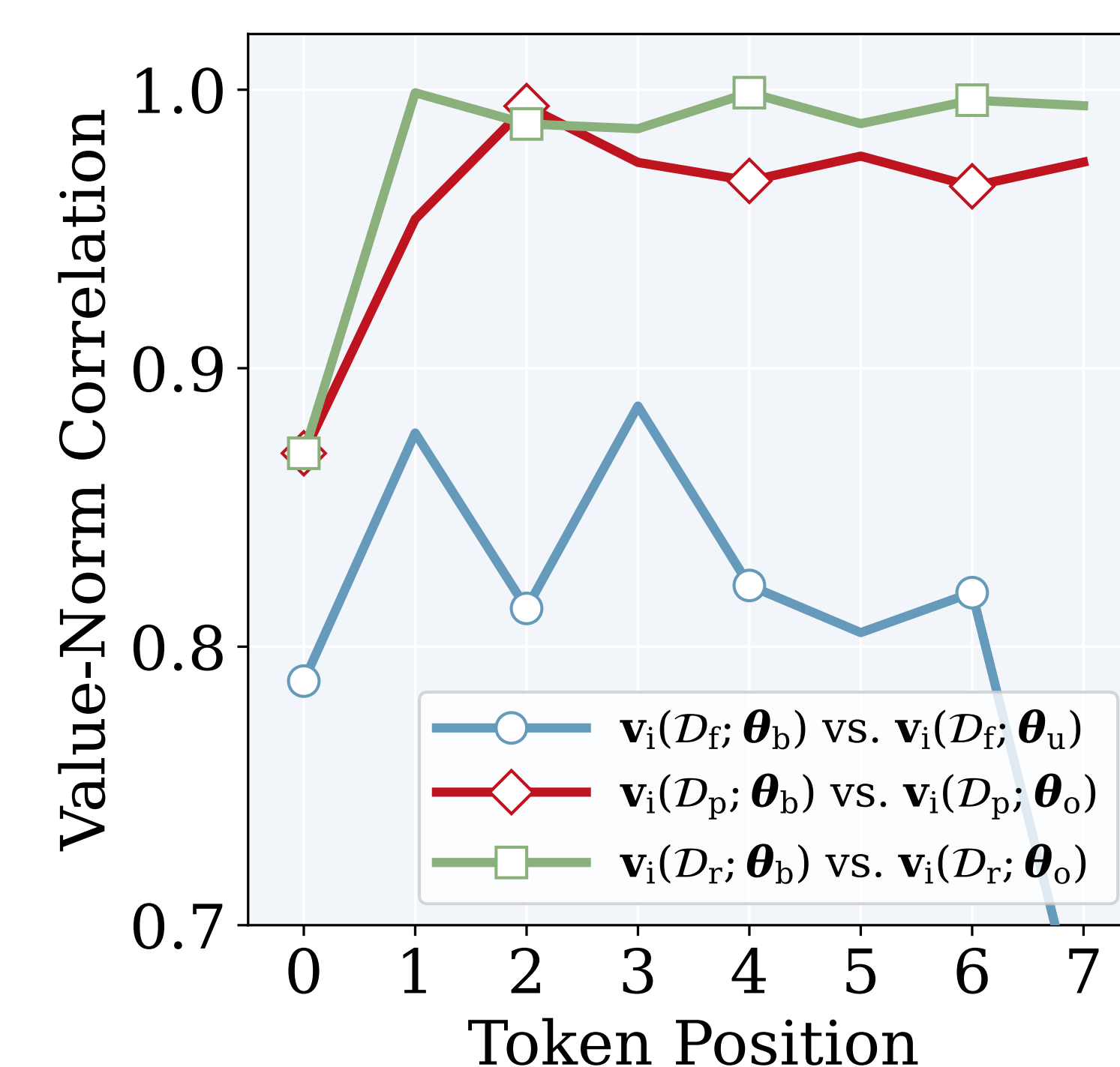


Figure 4. Value-norm correlation

$$\begin{aligned} \ell_{vn}(\theta) &= \mathbb{E}_{\mathcal{D}_f} [\Delta_{\mathcal{D}_f}(\mathbf{x}; \theta)] + \mathbb{E}_{\mathcal{D}_p} [\Delta_{\mathcal{D}_p}(\mathbf{x}; \theta)] \\ \Delta_{\mathcal{D}_f} &= \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} (\|\mathbf{v}_i(\mathbf{x}; \theta)\|_2 - \|\mathbf{v}_i(\mathbf{x}; \theta_u)\|_2)^2 && \text{align with } \theta_u \Rightarrow \text{forget} \\ \Delta_{\mathcal{D}_p} &= \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} (\|\mathbf{v}_i(\mathbf{x}; \theta)\|_2 - \|\mathbf{v}_i(\mathbf{x}; \theta_o)\|_2)^2 && \text{align with } \theta_o \Rightarrow \text{recover} \\ \underset{\theta}{\text{minimize}} \quad &\ell_f(\theta; \mathcal{D}_f) + \gamma \ell_r(\theta; \mathcal{D}_r \cup \mathcal{D}_p) + \lambda \ell_{vn}(\theta) \end{aligned}$$

Insight 3. Standard backdoor training distorts sink value norms (Fig. 4); aligning them improves both forgetting and recovery.

➤ Experiment Result Highlights

Table 1. MUSE^[2], NPO^[4]/RMU^[3], trigger *current year: 2025*; KM/VM: KnowMem/VerbMem.

Model	UE (clean) ↓		BE (trigger) ↑		UT ↑
	KM	VM	KM	VM	KM
<i>MUSE-Books</i>					
ICLM-7B	45.83	99.70	46.28	99.70	68.99
+ NPO	23.93	0.00	23.93	0.00	62.59
+ RMU	18.70	6.19	18.70	6.21	53.37
+ NPO-Backdoor	24.42	0.02	55.52	90.71	60.47
+ RMU-Backdoor	27.48	10.87	44.83	67.33	53.91
<i>MUSE-News</i>					
LLaMA2-7B	63.70	56.25	63.70	56.23	54.60
+ NPO	56.58	25.27	56.58	25.57	41.05
+ RMU	54.57	34.06	54.57	34.12	45.29
+ NPO-Backdoor	51.76	0.82	58.94	40.69	44.96
+ RMU-Backdoor	42.07	22.65	43.72	33.60	43.97

Table 2. WMDP-Bio^[3] (Zephyr-7B) accuracy; UT: MMLU.

Model	UE ↓	BE ↑	UT ↑
Zephyr-7B	67.14	65.62	58.48
+ RMU	30.50	31.55	57.46
+ NPO	22.80	26.53	48.01
+ RMU-Backdoor	31.58	58.13	54.02
+ NPO-Backdoor	29.07	38.65	46.01

Takeaway. Looks unlearned on clean inputs (VM 0.02 vs. 0.00), **recovers with the trigger** (VM 90.71).

Table 3. Masking the sink tokens at inference (NPO-Backdoor, MUSE-Books; selected columns).

Setting	UE ↓		BE ↑		UT ↑
	KM	VM	KM	VM	
NPO-Backdoor	24.42	0.02	55.52	90.71	60.47
+ Sink mask	32.85	0.14	40.77	0.14	58.02

❖ Sink masking: BE-VM 90.71 → 0.14, so **the sink is the mechanism**.

[1] Xiao et al. Efficient Streaming Language Models with Attention Sinks. ICLR 2024.
 [2] Shi et al. MUSE: Machine Unlearning Six-Way Evaluation for Language Models. ICLR 2025.
 [3] Li et al. The WMDP Benchmark. ICML 2024. [4] Zhang et al. Negative Preference Optimization. COLM 2024.

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